Non-Functional Requirements

Status: Requirements Baseline **Version:** 1.0 **Density:** Compact Spec

1. Security Requirements

ID	Requirement	Verification Method
NFR-SEC-01	Components shall use least-privilege IAM permissions.	Review IAM policies & role execution bounds.
NFR-SEC-02	Credentials, access keys, and certificates shall not be committed to Git.	Repository scan & gitignore review.
NFR-SEC-03	Automated response restricted to allowlisted test resources.	Test containment against out-of-scope targets.
NFR-SEC-04	Administrative and response actions shall leave auditable logs.	Inspect CloudWatch & audit log streams.
NFR-SEC-05	Data in transit and at rest shall use TLS and standard encryption.	Review MQTT TLS configs and DB encryption.
NFR-SEC-06	Input validation shall handle malformed events without system crash.	Fuzz test event parser with invalid JSON payloads.
NFR-SEC-07	Dashboard access protected by authentication mechanism.	Verify auth tokens and login controls.

2. Performance & Responsiveness

ID	Requirement	Target / Verification
NFR-PER-01	Dashboard renders primary view within target threshold.	< 3s load time across 10 sample runs.
NFR-PER-02	Event processing and risk scoring latency.	< 5s per test event (excluding AWS delays).
NFR-PER-03	Response execution latency from alert to action.	Log timestamp delta from trigger to isolation.
NFR-PER-04	Document tested scale (device count & throughput).	Report load test telemetry boundaries.

3. Reliability & Recovery

ID	Requirement	Verification Method
NFR-REL-01	Processing failures logged explicitly without silent success.	Simulate failed AWS API calls & verify errors.
NFR-REL-02	Distinguish state: Requested vs Attempted vs Confirmed.	Validate state transitions in incident records.
NFR-REL-03	Document procedures to restore isolated entities and clean lab.	Follow cleanup guide post-testing.
NFR-REL-04	End-to-end demonstration scenario stability.	3 consecutive successful test scenario runs.

4. Usability & Accessibility

ID	Requirement	Verification Method
NFR-USE-01	Clear severity, status, and outcome labeling.	UI review with peer tester.
NFR-USE-02	Risk scores must feature contributing factor breakdowns.	Inspect incident breakdown view.
NFR-USE-03	Workflow understandable without direct AWS console access.	Conduct demo solely via dashboard UI.
NFR-USE-04	Responsive interface across standard laptop resolutions.	Viewport testing at 1366x768 & 1920x1080.

5. Maintainability & Code Quality

ID	Requirement	Verification Method
NFR-MNT-01	Modular codebase (ingestion, engine, response, UI).	Code structure review.
NFR-MNT-02	Externalize configuration and risk thresholds.	Review config file / env variables.
NFR-MNT-03	Comprehensive README and execution instructions.	Clean environment setup test.
NFR-MNT-04	Version control discipline with clear commit histories.	Git commit log review.
NFR-MNT-05	Unit testing for core risk calculation engine.	Execute automated test suite.

6. Data Management & Observability

ID	Requirement	Verification Method
NFR-DAT-01	Synthetic or explicitly approved data for test scenarios.	Inspect test data fixtures.
NFR-OBS-01	Diagnostic logging for failures without credential exposure.	Inspect application error logs.
NFR-OBS-02	Incident timeline linking root cause events to actions.	Verify audit trail linkage.
NFR-OBS-03	Consistent UTC timestamp formatting across services.	Verify ISO 8601 timestamp compliance.

7. Cost & Operational Constraints

ID	Requirement	Verification Method
NFR-CST-01	Monitor and identify billable AWS services prior to activation.	Maintain active cost tracking sheet.
NFR-CST-02	Resource cleanup after test execution to avoid idle charges.	Resource teardown verification.
NFR-CST-03	Document cloud dependencies requiring paid service tiers.	Architecture prerequisite list.